Symmetry

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Symmetry Basics

1. 4

2. C) 3

3. 6

4. C) 2

5. 1

6. 2

7. D) A (H and O each have 2 lines; N and S have 0 lines)

8. 1

9. 8

10. A, C and E (square, rectangle and regular pentagon all have at least one line of symmetry; a scalene triangle and a non-rectangle/non-rhombus parallelogram have none)

11. 36 cm² (18 × 2)

12. (2, −7)

13. B) (−4, 3)

14. 2 (the two diagonals)

15. 1

Symmetry & Reflection

16. 4

17. 3

18. E) 6

19. 2

20. Order 5; 5 lines of symmetry

21. C) Parallelogram (not a rectangle or rhombus) — it maps onto itself after a 180° turn but has no mirror line

22. (2, 5)

23. (−3, −4)

24. A, B, D and E (H, N, S and Z each map onto themselves after a 180° turn; F does not)

25. (11, 1) (reflecting x = 1 in the mirror line x = 6 gives x = 2×6−1 = 11; the y-coordinate stays 1)

26. C) 2

27. 9 sides (a regular nonagon); order of rotational symmetry 9

28. (6, 4) (reflecting y = −2 in the line y = 1 gives y = 2×1−(−2) = 4)

29. Kite (it has exactly 1 line of symmetry and no rotational symmetry beyond order 1; the square, rectangle and regular pentagon all have more than one line and/or rotational symmetry greater than order 1)

30. D) Scalene triangle (it has 0 lines of symmetry and only order 1 rotational symmetry)

Advanced Symmetry Challenge

31. 8 sides; 8 lines of symmetry

32. B) Regular pentagon

33. (4, −5) (reflecting twice in the same mirror line returns the point to its original position)

© PassFrame Tuition — for registered student use

34. (6, 0) (reflecting x = 0 in the mirror line x = 3 gives x = 2×3−0 = 6; the y-coordinate stays 0)

35. A, B and D (the final image is (−4, −3), which is also what a 180° rotation about the origin gives, and its x-coordinate is negative; C is false since the final image is not (4, −3), and E is false since the y-coordinate, −3, is negative, not positive)

36. 36 cm^2 (rotations preserve area, so the image has the same area as the original shape)

37. D) 9

38. (7, 2) (reflecting x = −3 in the line x = 2 gives x = 2×2−(−3) = 7; the y-coordinate stays 2)

39. 4 lines (the two diagonals, the vertical line and the horizontal line all pass through the centre)

40. A) It has no rotational symmetry (it must be turned through a full 360° before it looks the same again)

41. 60° (360° ÷ 6 = 60°)

42. No, the student is not correct. For a regular shape, the order of rotational symmetry always equals the number of lines of symmetry, so a shape with 4 lines of symmetry (a square) has rotational symmetry of order 4, not 2.

43. B) Scalene triangle (a parallelogram and letter S both have order-2 rotational symmetry, and a rectangle and isosceles triangle both have at least 1 line of symmetry; only the scalene triangle has neither)

44. (2, −6) (reflecting (6,2) in y = x swaps the coordinates to give Q = (2, 6); reflecting Q in the x-axis keeps x the same and negates y, giving R = (2, −6))

45. A, B and C are true (D is false — a scalene triangle has no symmetry at all; E is false — a non-special parallelogram has rotational symmetry of order 2 but no lines of symmetry)

© PassFrame Tuition — for registered student use

Homework Worksheet — Answers

Symmetry Basics

1. 5

2. 1

3. 2

4. D) 8

5. 1

6. B) X (F, N, S and L each have 0 lines of symmetry)

7. 0

8. 1

9. A) 0

10. A, B and D (a non-rectangle/non-rhombus parallelogram, a scalene triangle and the letter N all have 0 lines of symmetry; a square has 4 and the letter T has 1)

11. (6, −1)

12. (3, 2)

13. 44 cm² (22 × 2)

14. A) (5, −5)

15. 6

Applied Symmetry

16. 8

17. 2

18. D) 5

19. 1

20. Order 7; 7 sides (a regular heptagon)

21. B) Letter Z (it maps onto itself after a 180° turn but has no mirror line)

22. (−3, 4)

23. (1, −2) (reflecting y = 6 in the line y = 2 gives y = 2×2−6 = −2)

24. A, B, D and E (O, Z, N and H each map onto themselves after a 180° turn; T does not)

25. (14, 3) (reflecting x = 2 in the mirror line x = 8 gives x = 2×8−2 = 14; the y-coordinate stays 3)

26. C) 2

27. 11 sides; order of rotational symmetry 11

28. (2, 5) (reflecting x = −4 in the line x = −1 gives x = 2×(−1)−(−4) = 2; the y-coordinate stays 5)

29. Square (a scalene triangle has neither; a parallelogram has rotational symmetry but no line symmetry; a kite has only 1 line and no rotational symmetry beyond order 1)

30. D) Letter N (it has rotational symmetry of order 2 but no line symmetry, so it is not an example of a shape with both)

Standard / Advanced

31. 4 lines (a regular hexagon has 6 lines of symmetry; 6 − 2 = 4)

32. 8 (a square has 4 lines of symmetry, so its order of rotational symmetry is also 4; doubled = 8)

33. (3, 6) (reflecting twice in the same mirror line returns every point to its original position)

34. 3 sides (a regular hexagon has 6 lines of symmetry; half of 6 is 3, and a regular polygon with rotational symmetry of order 3 is a triangle, which has 3 sides)

35. N (H and O both have line symmetry as well as order 2, and T has line symmetry but order 1; only N has order 2 and no lines of symmetry)

© PassFrame Tuition — for registered student use

36. No, it is not possible. For a genuine regular polygon, the number of lines of symmetry always equals its order of rotational symmetry (both equal the number of sides). A regular pentagon has 5 lines of symmetry, so its order of rotational symmetry must also be 5, not 8 — there is no regular polygon with 5 lines of symmetry and rotational order 8 at the same time.

37. A) (2, 5) (reflecting twice in the same mirror line returns the point to its original position)

38. 2 lines of symmetry and rotational symmetry of order 2 (any rectangle that is not a square always has exactly 2 lines of symmetry and order 2, regardless of how it was built)

39. C) 7 (a square has 4 lines of symmetry; 4 + 3 = 7, and a regular heptagon has rotational symmetry of order 7)

40. (3, −3) (reflecting (3,3) in y = x gives B = (3, 3), since the point lies on the mirror line itself; reflecting B in the x-axis gives C = (3, −3))

41. C) 8 lines (a square has order 4, so the second polygon has order 2×4 = 8, making it a regular octagon, which has 8 sides and therefore 8 lines of symmetry)

42. A, B, C and D (the origin and any point on y = x map to themselves, and the reflection rule swaps coordinates; D is correct since (4,−1) maps to (−1,4); E is false since the reflection swaps coordinates rather than changing their signs)

43. Each piece is a right-angled isosceles triangle; area of each piece = 9 cm^2 (the square's area is 6 × 6 = 36 cm^2, divided equally into 4 pieces)

44. C) 10 (n^2 = 100 means n = 10, so the polygon has 10 sides)

45. Reflecting P = (3, 5) in the x-axis gives (3, −5). Rotating (3, −5) by 180° about the origin gives Q = (−3, 5). Reflecting P directly in the y-axis gives R = (−3, 5). Q and R are the same point: (−3, 5).

© PassFrame Tuition — for registered student use